Harmonic Vector Equilibrium: A Unified Integer-Based Framework for Control, Navigation, and Communication on the Cuboctahedral Matrix

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***Abstract*—This disclosure presents an integer-based spatial and phase control framework designed for embedded systems and actuators. The method replaces conventional floating-point arithmetic and 360-degree or radian metrics with a base resolution of $L = 27720$. This specific metric, being the Least Common Multiple (LCM) of integers 1 through 12, guarantees exact integer division for complex mechanical gear ratios and spatial subdivisions. The framework eliminates progressive floating-point phase drift and provides a computationally lightweight, pure-integer control law suitable for resource-constrained microcontrollers. The complete C++ implementation and LUT generators are available as open-source prior art at: https://github.com/alexlozovoj1977-cmd/HVE-Phase-Controller.**

I. The Problem with Floating-Point Cartesian Control

Traditional actuator control systems (such as PID loops) heavily rely on floating-point arithmetic to compute phase, angle, and positional error. In continuous rotary systems, the continuous accumulation of positional data using `float32` or `float64` inevitably introduces quantization and rounding errors. Over time, this leads to accumulated phase drift. Furthermore, mapping continuous physical angles to standard digital bounds results in fractional values when interfacing with mechanical transmissions that use prime-number gear teeth counts.

II. The L=27720 Metric

To resolve these hardware-software interfacing inefficiencies, this framework introduces $L = 27720$ as the fundamental quantum of a full rotation ($360° \equiv 27720$ units).

A. Mechanical and Computational Justification

The metric 27720 is the LCM of $\{1, 2, \ldots, 12\}$. While it is the largest multiple fitting efficiently within signed 16-bit boundaries (`int16_t` bounds: ± 32767), by discretizing the control space into 27720 units, the system achieves:

1) **Exact Gear Ratio Mapping:** Mechanical reductions utilizing gears with 3, 7, 9, or 11 teeth can be mapped to exact integer values without fractional remainders.
2) **Bounded Quantization Error:** The quantization error is strictly bounded to $\pm 0.0065°$ (half a step) and does not accumulate over time, mathematically eliminating phase drift.
3) **Sufficient Granularity:** A single unit corresponds to $360/27720 \approx 0.013°$, which exceeds the mechanical backlash of most standard servo actuators.

III. Integer-Based Control Law Architecture

This work builds directly upon the MOTION controller [1], which established the bounded port-Hamiltonian formulation on the compact manifold S^1. Here, the control loop is formulated as a pure-integer phase controller operating on a discrete ring of size L.

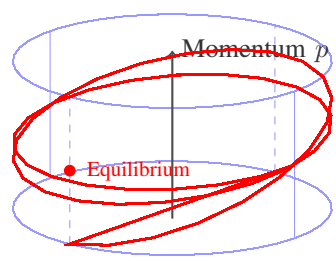

Figure 1. Phase portrait of the continuous counterpart on the cylindrical manifold. The $L = 27720$ metric discretizes this exact topology, mapping the ring into strictly integer states.

A. Event-Triggered Gain Scheduling

Instead of standard linear error multiplication, the system utilizes a sinusoidal lookup table (LUT) mapped to the L-metric. To ensure rapid target acquisition without destabilizing the system near the equilibrium point, the architecture employs event-triggered gain scheduling: the gain coefficient K dynamically scales based on the magnitude of the error, remaining strictly within the local contraction region of the generalized stiffness curve.

IV. Edge Cases and Spatial Topology

A. Saddle Point Singularity Resolution

In pure Cartesian systems, a target located exactly at 180° (or 13860 units) creates a state of indeterminate direction. This framework implements a deterministic override (`error = 13859`), breaking the unstable equilibrium at the cost of a deliberate, microscopic positional perturbation ($\approx 0.013°$). This systematic error is negligible for physical systems but critical for ensuring algorithmic liveness.

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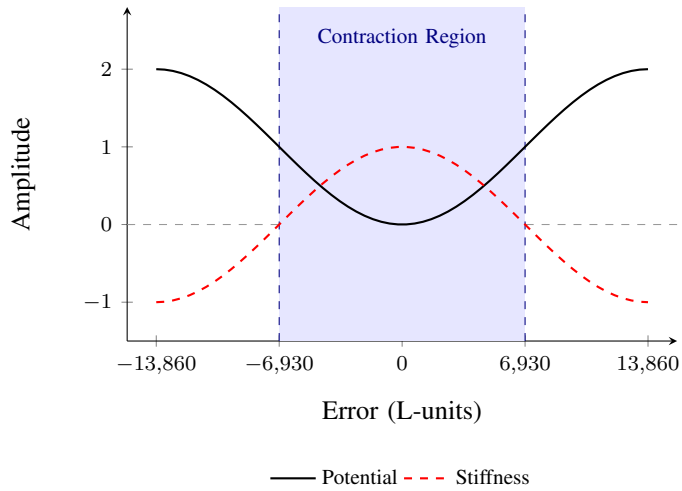

Figure 2. Local contraction region. Trajectory convergence is strictly guaranteed where generalized stiffness remains positive (Error $< |6930|$ units).

B. Cuboctahedral Integer Mapping

For 3D spatial navigation, target coordinates are mapped to the 12 primary vertex vectors of a cuboctahedron. The $L = 27720$ metric enables exact integer mapping of these vertices without floating-point representation (Azimuth is measured counterclockwise from the Front/X-axis, where V_1 and V_9 align at 0 units).

Table I
CUBOCTAHEDRON VERTICES IN $L = 27720$ SPHERICAL COORDINATES

Plane	**Vertex Index**	**Azimuth** (L)	**Elevation** (L)
Upper (+45°)	V_1 (Front-Right)	0	3465
Upper (+45°)	V_2 (Back-Right)	6930	3465
Upper (+45°)	V_3 (Back-Left)	13860	3465
Upper (+45°)	V_4 (Front-Left)	20790	3465
Equatorial (0°)	V_5 (Right)	3465	0
Equatorial (0°)	V_6 (Back)	10395	0
Equatorial (0°)	V_7 (Left)	17325	0
Equatorial (0°)	V_8 (Front)	24255	0
Lower (−45°)	V_9 (Front-Right)	0	-3465
Lower (−45°)	V_{10} (Back-Right)	6930	-3465
Lower (−45°)	V_{11} (Back-Left)	13860	-3465
Lower (−45°)	V_{12} (Front-Left)	20790	-3465

For intermediate coordinates, the framework utilizes an integer-weighted barycentric interpolation between adjacent vertices:

$$V_{target} = \frac{(L - W) \cdot V_A + W \cdot V_B}{L}$$

where $W \in [0, L]$ is an integer representing the arc length along the geodesic between the target and V_A.

V. GENERALIZED SCOPE AND CROSS-DISCIPLINARY APPLICATIONS

While the initial implementation targets 16-bit embedded actuators ($L = 27720$), the underlying mathematical principle—mapping physical phase spaces to exact Least Common Multiple (LCM) integer metrics—is inherently scalable to higher-precision systems (e.g., 32-bit LCM basis). This disclosure establishes prior art for employing this framework across multiple domains:

1) Medical Imaging and Neuromodulation: Utilizing the integer metric for drift-free spatial encoding in k-space during Magnetic Resonance Imaging (MRI), eliminating phase accumulation errors during long sequences. In non-invasive neuromodulation, it provides absolute phase-locking for multi-channel temporal interference to generate precise low-frequency beat envelopes (e.g., targeted resonance) without spatial focal drift.

2) High-Energy Physics and Plasma Confinement: Applying cuboctahedral phase control to stellarator and tokamak magnetic coils to dynamically compute confinement contours using pure integer math, mitigating numerical stochastic noise that contributes to plasma leakage in harmonic modulation systems.

3) Telecommunications: Employing Cuboctahedral Phase-Shift Keying, where 12 independent data channels are mathematically isolated using the exact integer divisibility of the L-metric, preventing intermodulation distortion.

VI. OPEN SOURCE IMPLEMENTATION

The core logic of the algorithm is demonstrated below. The complete architecture, including the full `sin_quarter_LUT` and simulation environment, is available at the project repository: https://github.com/alexlozovoj1977-cmd/HVE-Phase-Controller.

```
int32_t compute_harmonic_control(int32_t target_L, int32_t current_L, int32_t K_GAIN) {
    int32_t error = target_L - current_L;

    // Shortest path calculation on the ring manifold
    if (error > L_HALF) {
        error -= L_METRIC;
    } else if (error < -L_HALF) {
        error += L_METRIC;
    } else if (error == L_HALF || error == -L_HALF) {
        error = L_HALF - 1; // Singularity override
    }

    // Harmonic coupling via quarter-wave symmetry resolver
    int16_t sin_val = fast_sin_L(error);

    // Divide by 32768 (>> 15) to normalize Q15 LUT output scale
    return (K_GAIN * (int32_t)sin_val) >> 15;
}
```

Listing 1. Core Integer Phase Controller Logic

VII. ACKNOWLEDGMENTS

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REFERENCES

[1] O. Lozovyi, "MOTION: A Bounded Nonlinear Phase Controller Formulated as a Port-Hamiltonian System on Compact Manifolds," *Technical Disclosure Commons*, dpubs_series/9713, 2026.